Cost Effective Housing Techniques

Unit V: Housing in Disaster Prone Areas

Syllabus

Housing in disaster prone areas: introduction - earthquake - damages to houses - traditional / disaster prone areas - types of damages and failures of non-engineered buildings - repair and restoration of earthquake damaged non-engineered buildings - recommendations for future constructions - requirements of structural safety of thin precast roofing units against earthquake forces - status of R&D; in earthquake strengthening measures - floods, cyclone, future safety.

1. Introduction

India is one of the **most disaster-prone countries**. About **59% of the land is earthquake prone, 12% flood prone (40 million hectares), 8% cyclone prone** (7500 km coastline) and 68% of the area is drought prone. Disasters kill thousands, destroy lakhs of houses and push the poor back into poverty. **Houses (especially non-engineered) are the main source of loss of life**, so **safe housing** is the best disaster protection: "**Earthquakes do not kill people, buildings do.**"

Hazard	Housing impact
Earthquake	Collapse of walls and roofs, killing people inside
Flood	Submergence, erosion of foundations, collapse of mud walls
Cyclone / storm surge	Roof blown off, wall collapse, flooding of coastal houses
Landslide	Burial of houses in hills
Fire, drought, tsunami (2004)	Loss of houses, services, livelihoods

- **Disaster management cycle:** prevention - mitigation - preparedness - response - relief - **recovery (reconstruction)** - development; **Disaster Management Act 2005**, **NDMA** (National Disaster Management Authority) and **State SDMAs**; "**Build Back Better**" is the approach for reconstruction
- **Vulnerability Atlas of India (BMTPC, 1997, 2006, 2019)** shows **hazard-wise vulnerability of housing for each district** (wind, flood, seismic, landslide) and the

number of houses at risk

2. Earthquake: Basic Concepts

An **earthquake** is sudden shaking of the ground caused by the release of **strain energy** stored in rocks (along **faults**, plate boundaries). The Indian plate moves north and collides with the Eurasian plate: it causes the **Himalayan** earthquakes.

Term	Meaning
Focus (hypocentre)	Point under the ground where rupture starts
Epicentre	Point on the surface directly above the focus
Magnitude	Energy released (**Richter scale**, log scale; each unit = 32 times energy)
Intensity	Effect on structures and people at a place (**MSK / Modified Mercalli, 12 grades**); decreases with distance
Seismic waves	**P waves (compression, fast), S waves (shear, damaging), surface waves (Love, Rayleigh)**
Seismograph / accelerograph	Records ground motion; **PGA (peak ground acceleration)**
Liquefaction	Loose saturated sand loses strength: buildings tilt and sink
Tsunami	Sea waves due to sea-bed earthquake

Seismic zones of India (IS 1893 (Part 1): 2016)

Zone	Zone factor Z	Severity	Areas (examples)
V	0.36	Very severe	**Kashmir, NE India, Rann of Kutch, Andaman**
IV	0.24	Severe	**Delhi, Bihar, Uttarakhand, Sikkim, parts of Gujarat and Maharashtra coast**
III	0.16	Moderate	**Mumbai, Chennai, Kerala, parts of WB, Odisha, Rajasthan**
II	0.10	Low	**Most of the peninsular India: Hyderabad, Bengaluru (in II)**

- Zone VI was removed from the 2002 and later revisions; **zone map has only 4 zones (II to V)**
- **Notable earthquakes: Latur (Killari) 1993 (M 6.2), Uttarkashi 1991, Chamoli 1999, Bhuj 2001 (M 7.7, 13,800 deaths), Kashmir 2005 (M 7.6), Sikkim 2011, Nepal 2015** - most damage to **non-engineered** masonry and mud houses

Seismic design force (IS 1893)

Design horizontal seismic coefficient: $A_h = (Z/2) \times (I/R) \times (S_a/g)$

Design base shear: $V_B = A_h \times W$ (W = seismic weight of the building)

Z = zone factor, I = importance factor (1.0 / 1.5), R = response reduction factor (1.5 for unreinforced masonry, 3-5 for RC frames), S_a/g = spectral acceleration

Example

A one-storey masonry house, W = 600 kN, Zone IV (Z = 0.24), I = 1, R = 1.5, S_a/g = 2.5.

$A_h = (0.24/2) \times (1/1.5) \times 2.5 = 0.20$; $V_B = 0.20 \times 600 =$ **120 kN** (20% of the weight acts horizontally!) - hence **light roofs** are important.

3. Damages to Houses in Earthquakes

Earthquake force is an **inertia force = mass x acceleration**: heavier building, higher force. Walls and roofs shaken **in-plane and out-of-plane**. **Non-engineered buildings** (built without engineer, by owner or village mason with local materials: **mud, adobe, stone, brick in mud, low-strength masonry**) form **majority of Indian housing** and cause the maximum deaths.

Disaster-prone areas (regional vulnerability)

Region / Hazard	Typical vulnerable houses
Himalayan belt, NE India (seismic IV-V)	**Stone and mud masonry, brick in mud, heavy slate / mud roofs**
Gujarat, Kutch	Stone and brick houses, **heavy RCC / stone slab roofs**, bhungas (good)
Indo-Gangetic plain (Bihar, UP) - seismic and flood	**Mud / adobe, kachcha houses**
East and West coasts - cyclone, storm surge	Kachcha, thatch, weak masonry
Brahmaputra and Ganga basins - floods	Mud and bamboo houses
Central India (Latur) - moderate seismic but very vulnerable houses	**Heavy stone masonry with mud mortar and heavy mud roofs**

4. Types of Damages and Failures of Non-Engineered Buildings

Failure type	Description and cause
Diagonal (X) shear cracks in walls	Due to **in-plane shear**; around openings; weak mortar, long walls
Separation of walls at corners and T-junctions	Walls vibrate in different directions; no **corner stitching / bands**; vertical crack at corners
Out-of-plane collapse of walls (overturning)	Walls too tall, thin, slender, unsupported by cross walls or roof; **mud mortar**
Gable wall failure	Triangular gable not tied to roof
Collapse of roof / floor	**Heavy roofs (mud phuska, stone slab) fall; no connection of roof to walls; insufficient bearing**
Delamination of walls (wythes separation)	Two-leaf stone walls, no through stones
Damage around openings	Large openings near corners, **no lintel band**, door-window cracks
Parapet, chimney, balcony, water tank, cornice falls	Cantilevers and projections
Foundation failure, settlement, liquefaction	Poor soil, no plinth band; tilted houses (Bhuj, Kutch)
Pounding between buildings	Adjacent buildings of different heights without gap
Soft storey, short column, torsion (RC buildings)	Open ground floor for parking, irregular plan
Shattering of precast roofing / slippage of units	Inadequate bearing, no connection

Causes of poor performance

* **Heavy roof** and walls (large mass), **weak materials** (mud, lime mortar, random rubble, poor bricks), **poor workmanship**
* **No horizontal bands, no vertical bars at corners, long slender walls**, unsymmetrical plan, **big openings**
* **Lack of enforcement of codes**, no engineers, **ignorance**, and **weak connection of parts** (wall-wall, wall-roof)

5. Repair and Restoration of Earthquake-Damaged Non-Engineered Buildings

After the earthquake, a **damage assessment** by trained teams classifies the buildings (e.g., **Green – safe, Yellow – restricted use, Red – unsafe**) before deciding on **demolition, repair or strengthening**.

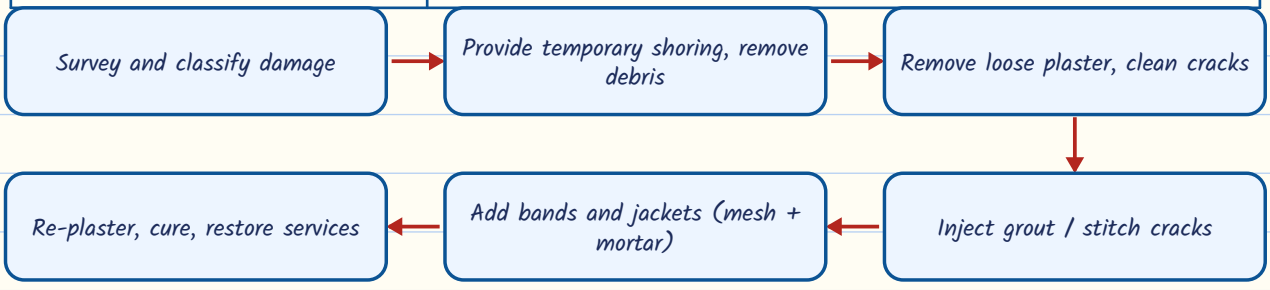
Damage grade (EMS)	Condition	Action
G1 Negligible / slight	Hairline cracks, small plaster falls	Cosmetic repair
G2 Moderate	Cracks in many walls, some plaster fall	**Repair + strengthening (non-structural / structural)**
G3 Substantial / heavy	Large wide cracks, wall failure, tilting	**Major structural repair and retrofit**
G4 Very heavy	Serious wall failure, partial roof collapse	Demolish or major reconstruction
G5 Destruction	Total or near collapse	Demolish and **rebuild** (with seismic design)

Terms

- **Restoration (repair):** bring the damaged building back to the **original strength**
- **Seismic strengthening / retrofitting:** upgrade strength **beyond the original** by adding new elements to resist future earthquakes
- **Rehabilitation:** repair + strengthening + services restored; **reconstruction:** new construction after demolition

Repair techniques (IS 13935: Repair and seismic strengthening of buildings)

Technique	Description
Crack filling / grouting	Clean cracks; **cement-sand grout / lime grout / epoxy injection**; **cement slurry for cracks 3-5 mm**, epoxy resin for cracks less than 3 mm; hairline cracks by **cement paste**
Stitching of wide cracks	**U-shaped steel dowels / stitching bars (6-8 mm) in grooves** across cracks, grouted with cement mortar
Re-building of badly damaged wall portions	Dismantle and re-build with **cement mortar**, **toothed joints**
Splints and bandages	**Reinforced concrete (or ferrocement) belts / bands** around the house at plinth, sill, lintel and eaves levels, with **vertical splints** at corners and junctions (bars in grooves, then plaster) **(2 or 3 per wall)**
Jacketing (ferrocement / mesh with mortar)	**Welded wire mesh (50 x 50 mm, 3-4 mm wire) fixed on both faces of wall by cross-ties through the wall, then plastered with 1:3 cement mortar (30-40 mm)**; **very effective and cheap**
Shotcrete / gunite overlay	Sprayed concrete over mesh
Grouting of walls and replacement of mortar	**Injection of cement-sand or lime grout** to strengthen weak rubble walls; **re-pointing**
Strengthening of foundation, roof connections	**Anchoring the roof** to walls through **bolts, wire ties; replace the heavy roof by a light roof**
Strengthening of openings	**Frames of steel angles** around big openings; fill unneeded openings
Steel belts / tie rods / buttresses	External supports, **corner buttresses**
Fibre reinforced polymers (FRP) wrapping	Carbon / glass fibre strips; higher cost



- **Precautions: materials (cement mortar not mud), hold-fasts / cross ties spaced 500-600 mm**, curing 7-14 days, **work by trained masons**, use of **local materials**, **owner participation**; **do not re-use the broken heavy roof**

- **Cost:** repair and strengthening usually **30–50% (or 10–30%)** of reconstruction cost; saves resources and time

6. Recommendations for Future Constructions

IS 4326:2013 (Earthquake resistant design and construction of buildings: Code of practice), **IS 13827** (Earthen buildings), **IS 13828** (Low strength masonry buildings), **IS 13920** (Ductile detailing of RC), **IS 1893** (design criteria), **IS 15988** (assessment and retrofitting of RC buildings) and **NDMA guidelines**.

General principles

- **Simple, symmetrical, regular plan** (square or rectangular, **L / B not more than 3**); avoid **L, T, U shapes** - separate them by **seismic joints (gap 25–40 mm)**
- **Light roof and light walls**; **low height** (1–2 storeys for masonry; **storey height 2.7–3.2 m**; **mud buildings single storey** only)
- **Box action:** all walls and roof **tied together** by **horizontal bands and vertical bars**
- **Good foundation** on uniform firm soil; **no foundation on liquefiable / filled soil**; plinth band
- **Quality of material and workmanship**, **cement or cement–lime mortar (1:6)** in zones III–V; **proper bonding of masonry (header and stretcher)**, **through stones** in stone masonry (every 1.2 m), **toothed joints**

Provision (IS 4326 / 13828 / 13827)	Requirement
Wall thickness	Brick: min **230 mm**; stone: **350–450 mm**; **earthen walls: 300 mm minimum**
Mortar	Cement-sand **1:6** (zones III–IV), **1:4** (zone V) for brick; lime-cement mud only for low-rise **earthen buildings with bands**
Openings	**Total width not more than 50% of wall length (single storey), 42% (two storey)**; **min 600 mm from corner**, pier width not less than **340 mm**, **door–window up to 1/3 length in low strength masonry**
Horizontal bands	**Plinth, lintel, sill (optional), roof / eaves (gable) level**; RCC band **75–150 mm thick, 2 bars of 8–12 mm**, stirrups 6 mm @ 150 mm; **timber or bamboo bands in low cost houses**
Vertical bars	**At corners and T junctions**, jambs of doors: **8–12 mm bars (or bamboo) from foundation to roof**, in zone IV–V
Wall height / length	Wall height **max 8 x thickness (mud: 6–8 x)**; **length between cross walls max 6 m (4 m for mud)**

Provision (IS 4326 / 13828 / 13827)	Requirement
Roof	**Light** (GI sheets, tiles, thatch on timber / bamboo trusses), **anchored with ties**; avoid heavy mud / stone slabs
Plinth and foundation	Depth 450–750 mm, width 600–900 mm in stone / brick; **continuous**

Seismic features of a low-rise masonry house

* **Roof band + lintel band + plinth band** make the house a rigid box
* **Corner reinforcement (steel / bamboo)** and **jamb bars, gable band**
* **Light roof tied** with bolts; **small openings**; **symmetry**; **mortar strong**

7. Safety of Thin Precast Roofing Units against Earthquake Forces

Thin **precast roofing / flooring units** (planks, joists, channels, cored units, ferrocement elements) are popular for low cost housing (Unit II), but they **can fall off** the walls in earthquake if not **connected** properly. The roof must work as a **rigid horizontal diaphragm**, transferring the **inertia forces** to the walls.

Requirement	Details
Adequate bearing	**At least 75 – 100 mm** on the supporting wall / bond beam (units 3 m or less span); prevents **slipping and falling**
Connection of units to supports	**Dowels, anchor bars, cast-in-situ RCC bond beam on the walls**; **hooks and bolts**; mortar bed 12 mm
Integral action of units	**Shear keys between adjacent units**, filling the joints with **1:3 mortar / micro-concrete, cast-in-situ screed (40–50 mm) with welded mesh or 6 mm bars @ 150–200 mm** acts as a **diaphragm**
Ring (bond) beam at roof level	**RCC / reinforced brick band all round**, connected to the wall steel; ties for all units
Light weight	Units, screed and finishes should be light (reduces the seismic mass and lateral force)
In-plane shear capacity	Joints and screed must carry the horizontal force $H = A_h \times W_{roof}$ through the **shear friction** to the wall bands
Vertical acceleration check	Units must resist **uplift (2/3 of horizontal)**, secure with ties or dowels
Stability of individual units	Ribs / joists should not tilt: **strutting and cross ties** before screed
Supports, openings	No large cut-outs without trimmer beams; **edges stiffened**

Requirement	Details
Quality control	Concrete cube strength **M20 or higher**, proof test of units, **IS 13990 (Precast RC planks and joists for roofing and flooring)**, **IS 15916 (precast buildings)**

Check for the roof connection

Horizontal seismic force on roof = A_h W. For roof weight 150 kN, A_h = 0.15 => H = 22.5 kN to be carried through the perimeter band to the walls. With roof perimeter = 30 m, shear per metre = 0.75 kN/m: dowels and screed should be designed for this shear (easily taken by a **mesh reinforced screed** and the **bond beam**).

8. Status of R&D; in Earthquake Strengthening Measures

- **Shake table tests** of **full-scale masonry houses** (CBRI Roorkee, **IIT Roorkee, IIT Kanpur, IIT Madras, SERC Ghaziabad**) show the effect of **bands, vertical bars, bamboo / timber reinforcement and mesh jackets**
- **Retrofitting techniques** developed: **ferrocement / wire mesh jackets, shotcrete, fibre wraps, steel belts, grouting, buttresses, external reinforcement; low cost seismic retrofitting of stone houses** (Latur, Kutch)
- **Low-cost seismic isolation** for low-rise houses (rubber pads, sliding base layers like **sand or graphite**) and **energy absorbing devices, dampers** are under research
- **New materials: bamboo and jute geo-grids, polypropylene (PP) bands** for earthen houses (**PP-band mesh in mud houses** tested by IIT Roorkee), **engineered bamboo, fibre-reinforced mortars**
- **Codes and guidelines:** BIS codes (IS 4326, 13827, 13828, 13935, 15988), **NDMA retrofitting guidelines, BMTPC Vulnerability Atlas, Guidelines for Improving Earthquake Resistance of Housing (NDMA)**
- **Awareness and training: mason training, Model house demonstration, Building Centres, Disaster Resistant Technologies in PMAY-G, Early Warning and seismic microzonation** (cities)
- **Gaps:** poor enforcement, **no mandatory approval in villages**, cost, lack of skilled people, low insurance coverage

9. Floods

Floods are caused by heavy rain, river overflow, dam breaks, drainage congestion and storm surges. Major flood zones: **Brahmaputra, Ganga, Kosi (Bihar), Mahanadi, Godavari and Krishna deltas; Assam, Bihar, UP, West Bengal, Odisha.**

Flood resistant construction

Measure	Detail
Site selection	Away from rivers, **above High Flood Level (HFL); flood plain zoning**
Raised plinth / platform	**Plinth above HFL (at least 0.6–1.0 m)** on **raised earth mounds (flood-proof platforms, villages on mounds); stilt houses on piles / bamboo posts** (Assam)
Foundation	**Deep, with scour protection; stone / brick with RCC, pile / pier foundations**; protect with **stone pitching**
Materials	**Burnt brick / stone / RCC, stabilised soil, waterproof plaster; no mud walls in flood area; bamboo, timber treated**
Services	Electrical points high up; **raised platform for toilets and hand pumps; sealed septic tanks**
Shelter	**Multi-purpose flood shelters on stilts, with a school / community**
Flood-proof houses	**Floating houses** (amphibious), **houses with openings to allow water to pass (wet-proofing)**
Drainage	Proper drains and **embankments, early warning**

10. Cyclones

Cyclones bring **strong winds (up to 200+ km/h)**, heavy rain and **storm surge** along the east coast (Andhra, Odisha, Tamil Nadu, Bengal) and Gujarat. Houses are damaged by **wind pressure and suction on the roof**, flying debris and flooding.

Wind pressure (IS 875 Part 3)

$V_z = V_b \, k_1 \, k_2 \, k_3 \, k_4$ (design wind speed; V_b = basic wind speed: 33 - 55 m/s)

$p_z = 0.6 \, V_z^2$ (N/m^2); Force = $(C_{pe} - C_{pi}) \, A \, p_z$

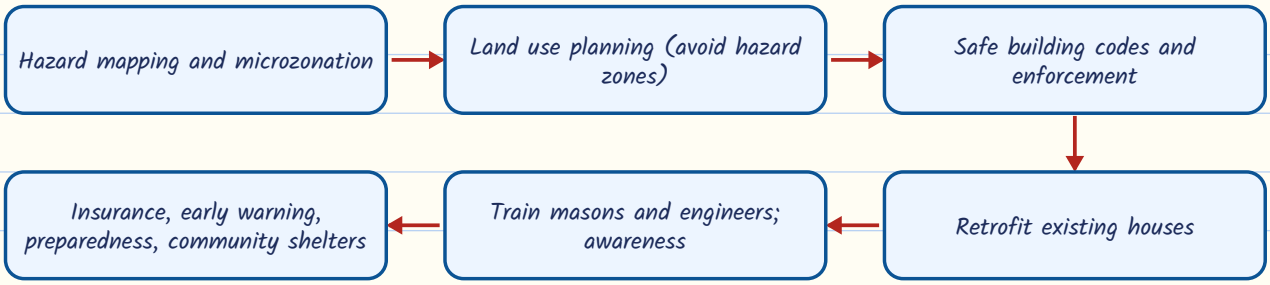
Example: V_z = 50 m/s => $p_z = 0.6 \times 2500 = 1500$ N/m^2 (1.5 kN/m^2); roof suction may be 1.0 - 1.5 times this value.

- **Cyclone resistant design: compact, square / octagonal plan, hip roof with slope 25–30 degrees** (gable ends are weak), **small eave projection (not more than 500 mm)**, low height; **strong roof-to-wall connection: tie-down bolts, straps,**

"**hold-fasts**" to the plinth beam or foundation; **strong cross bracing of roof; fix GI sheets with J-bolts and washers; avoid large openings on windward side; shutters**

- **Foundation** deep enough, **RCC columns and walls, brick / concrete walls rather than mud; plant wind-breaks (trees)** (shelter belts), **coastal embankments**
- **Cyclone shelters** (multi-purpose: raised, strong, with water, sanitation) - **about 1 per 5000 people** along coast; **early warning (IMD) and evacuation**
- **Storm surge:** houses on **raised plinth / platforms, mangrove belt**, avoid construction within the coastal regulation zone (**CRZ**)

11. Future Safety: Mitigation and Planning



- **Land use zoning and regulation**: no development in floodways, landslide slopes, coastal hazard zones; set backs
- **Compulsory use of codes (BIS, NBC 2016), techno-legal regime, third-party checks, licensed engineers and masons; disaster resistant features in all government housing (PMAY, IAY)**
- **Retrofitting** of critical buildings: schools, hospitals, shelters; **housing insurance** (e.g., **National Agricultural / rural Housing insurance**), **disaster risk transfer**
- **Community-based disaster preparedness, mock drills, training "Aapda Mitra", community participation, women involvement**
- **Use of low cost technology**: stabilised blocks, bands, bamboo, ferrocement, **light roofs**, and **owner-driven reconstruction** with technical help; **research and development, information systems (GIS, remote sensing)**

Remember for exam

* Seismic zones: II (0.10), III (0.16), IV (0.24), V (0.36); 59% of India is earthquake prone
* $V_B = A_h W$, with $A_h = (Z/2)(I/R)(S_a/g)$; heavy roof = heavy force

* Failures: diagonal cracks, corner separation, out-of-plane collapse, roof fall, gable failure
* Repair: grouting, stitching, splints and bandages, mesh jacketing; bands + vertical bars + light roof for new houses
* Precast units: bearing 75-100 mm, bond beam, screed with mesh, shear keys, good ties
* Floods: raise plinth above HFL, stilts; Cyclone: hip roof 25-30 degrees, tie-downs, shelters

Important questions

1. Explain the causes and damages to houses in an earthquake. Classify the seismic zones of India.
2. Describe the types of damages and failures of non-engineered buildings.
3. Explain the repair and restoration techniques for earthquake-damaged masonry buildings.
4. What are the recommendations for future construction in seismic areas (IS 4326)?
5. Explain the requirements of structural safety of thin precast roofing units against earthquake forces.
6. Discuss the status of R&D; in earthquake strengthening measures.
7. Describe flood and cyclone resistant features in housing.
8. Explain the future safety measures for housing in disaster prone areas.